Linear or not? The Keeling Curve — Project Report

DSM-101 / DSM-102 · Project 4 · Nov 10, 2026 · @Azwad Islam Sachcha

1. Question and hypothesis

CO_2 at Mauna Loa is linear only locally. Within one decade a straight line fits, but over 1958–2015 the growth rate almost triples, and a yearly cycle of about 6 ppm sits on top. We ask in which sense each model is linear, and how far a model of the past can be trusted for 2035.

Hypothesis, written before fitting: one straight line leaves a U-shape and a yearly wave in its residuals. A polynomial trend plus sines and cosines removes both while staying linear in its coefficients. An exponential trend $c_0 + a \cdot e^{(rt)}$ beats the polynomial on the held-out years. No model of the past knows future emissions, so the 2035 error bar must come from how past forecasts failed.

2. Data

We use NOAA GML's monthly mean file `co2_mm_mlo.txt`, kept unchanged in `data/raw/`. It covers March 1958 to September 2026 (823 months), in ppm, a mole fraction in dry air. The source, download date and citation are in `data/README.md`. We model the `average` column and use `deseasonalized` only as a check.

We keep the 194 **Scripps months** (Mar 1958 – Apr 1974), which have no day count or st.dev in the file. We flag Dec 1975 as **interpolated**, because it has no measured days; Apr 1984 has 2 measured days, so it counts as measured. We keep the 8 **Maunakea months** (Dec 2022 – Jul 2023), flagged and scored separately.

Split. We fit on Mar 1958 – Dec 2015 (694 months) and hold out Jan 2016 – Sep 2026 (129 months). We did not look at the held-out months until every model was fixed. For model selection we fit on 1958–2010 and validate on 2011–2015. Time t is in years since March 1958.

3. Rung 1 — one straight line

Least squares on all 694 training months gives

> $y(t) = 307.51 \text{ ppm} + 1.514 \text{ ppm/yr} \cdot t$

The model is affine in t and linear in its two parameters. The slope is the average growth rate over 58 years. The intercept is 8.2 ppm below the 315.71 ppm measured in March 1958. Training RMSE is 3.74 ppm; held-out RMSE is 13.74 ppm.

The residuals (`rung1_linear_baseline.png`) show two patterns. The first is a **U-shape**: positive at both ends and negative in the middle, so the trend curves upward. The second is a **yearly wave** of about ±3 ppm, clearest in 1983–1987, where the U is flat. Both are the first evidence of non-linearity.

4. Rung 2 — a straight line per decade

Within one decade CO_2 is close to a straight line, so we fit $y = \alpha + \beta \cdot t$ to each calendar decade separately (`rung2_decade_slopes.png`). The slope β is that decade's growth rate. It is **0.77, 1.19, 1.56, 1.53, 1.99 and 2.10 ppm/yr** for the 1960s to the 2010s; the 2010s cover only 2010–2015 (72 months). The growth rate has almost tripled.

The 22 months of 1958–1959 are skipped. They cover less than two yearly cycles, so their slope of −0.39 ppm/yr shows the season, not falling CO_2. The rising slope is why one straight line fails.

5. Rung 3 — polynomial trend plus harmonics

We add powers of t, and sines and cosines with periods of 1, 1/2, 1/3 … years:

> $y(t) = \Sigma_k \beta_k \cdot t^k \ (k = 0 \dots d) + \Sigma_j [a_j \cdot \sin(2\pi j t) + b_j \cdot \cos(2\pi j t)] \ (j = 1 \dots h)$

The curve bends in t, but every unknown multiplies a known column of the design matrix X. So $y = X\theta$ is linear in θ, and one least-squares solve gives the exact optimum. For numerical stability, the trend is fitted in $x = t/10$.

How many terms. We tried degree 1–8 with 0–12 harmonics, each fitted on 1958–2010 and scored on 2011–2015 (`rung3_selection_heatmap.png`). The minimum is **degree 2 with 4 harmonics**: 11 parameters, validation RMSE 0.504 ppm. Training error would have chosen differently: degree 3 lowers training RMSE (0.725 vs 0.756 ppm) but raises validation RMSE to 1.318 ppm. Two harmonics give 0.511 ppm, so the last two add little.

Refitted on 1958–2015, the trend is $314.22 + 0.8127 \cdot t + 0.01217 \cdot t^2$ ppm. The yearly cycle has an amplitude of 2.81 ppm, the half-year cycle 0.78 ppm. Training RMSE is 0.735 ppm, but held-out RMSE is 2.20 ppm: the parabola grows too fast after 2015.

6. Rung 4 — exponential trend, r fitted non-linearly

We replace the parabola with an exponential and keep the 4 harmonics, so both models have 11 parameters:

> $y(t) = c_0 + a\cdot(e^{(rt)} - 1) + \Sigma_j\,[a_j\cdot\sin(2\pi jt) + b_j\cdot\cos(2\pi jt)]$ $(j = 1 \ldots 4)$

The rate r sits inside the exponential, so the model is truly non-linear in r. For any fixed r it is linear in the other 10 coefficients. So we search r alone (`minimize_scalar`, bounded to 0.005–0.1 per year), with least squares inside every step. Training RMSE plotted against r has a single minimum (`rung4_rate_scan.png`).

Fit: r = 0.01590 per year, a = 58.95 ppm, c_0 = 313.55 ppm; the trend now grows at 2.79 ppm/yr. Both models fit 1958–2015 about equally well (training RMSE 0.796 vs 0.735 ppm). On the held-out years the exponential is three times better (0.76 vs 2.20 ppm). Its trend is also closer to NOAA's deseasonalised series there (RMSE 0.71 vs 2.19 ppm). For 2035 the two models disagree by 6.4 ppm: 456.1 against 449.7 ppm.

We trust the exponential for two reasons. It wins on the years neither model saw. It also has a mechanism: emissions have grown by a roughly constant percentage each year. The catch, shown in rung 5, is that r needs a long history to be pinned down.

7. Model comparison

All models were fitted on 1958–2015 and scored on the 129 held-out months. Errors are in ppm.

Model	Linear in	Params	Train RMSE	Held-out MAE	Held-out RMSE	Held-out SD
Exponential + seasonal	10 of 11 (not r)	11	0.796	0.645	**0.762**	0.550
Polynomial (d = 2) + seasonal	all 11	11	0.735	2.073	2.202	0.747
Linear	both 2	2	3.742	13.202	13.739	3.819

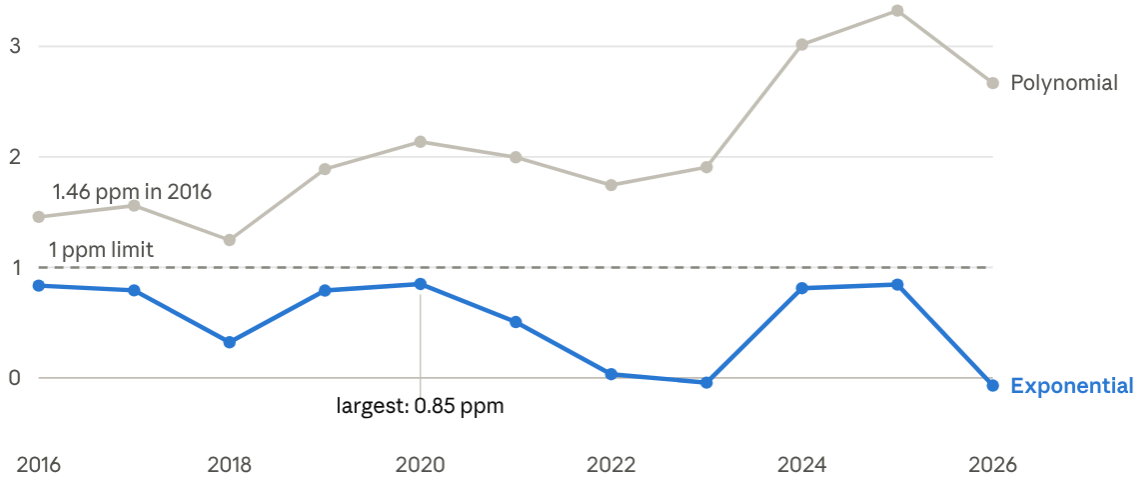
The SD is the spread once the mean bias is removed. The polynomial's SD is 0.75 ppm against an RMSE of 2.20 ppm, so its error is mostly a steady overshoot. On the 8 Maunakea months, held-out RMSE is no worse than on the rest (exponential 0.42 vs 0.78 ppm), so keeping them was safe.

8. Rung 5 — limits

How far ahead? Month by month, every model passes 1 ppm within 0–3 months because of the yearly wiggle and noise, so we read the horizon on annual means. There the exponential never passes 1 ppm in 2016–2026; its largest error is 0.85 ppm. The polynomial passes 1 ppm in the first held-out year, and the line is 8.6 ppm off from the start.

The exponential stays under 1 ppm in every held-out year; the polynomial never does

Annual mean error, model minus measured (ppm), models fitted on 1958 to 2015. The linear model is off the scale (over 8 ppm).



results/tables/rung5_annual_mean_error.csv · held-out years 2016–2026 (2026 holds 9 months)

The failure case. Fitted only on 1958–1985, the exponential predicts 401.5 ppm for 2005, against 380.0 ppm measured: 21.5 ppm too high (`rung5_failure_case.png`). The polynomial is 9.7 ppm too high and the line 13.1 ppm too low. Over 27 years the record is nearly straight, so many rates fit almost equally well. The fitted r falls from 0.047 per year (data to 1975) to 0.037 (1985), 0.020 (1995) and 0.016 (2015). A truly non-linear parameter is only as good as the range of data that constrains it.

2035 with an error bar. Monthly residuals are strongly correlated, so a least-squares interval would be too narrow. Instead we refit each model on data ending in 1985 … 1995, forecast the annual mean 20 years ahead, and take the 90th percentile of the absolute error. Twenty years is also how far 2035 lies beyond 2015.

The 2035 annual means are: **exponential 456.1 ± 21.5 ppm**, polynomial 449.7 ± 10.3 ppm and linear 424.5 ± 15.8 ppm. The exponential's bar is conservative. Its backtest error falls from 21.5 ppm (1985 cutoff) to 6.8 ppm (1995) as it sees more history, and our real fit has 20 more years than that. The linear forecast is already ruled out: it is below the 426.07 ppm measured in September 2026.

What no model of the past can know: future emissions (policy, the economy, a pandemic year), whether oceans and forests keep absorbing about half of them, and when El Niño years and eruptions come. The error bar measures how wrong extrapolation has been, not how the world will change.

9. Software and reproducibility

`python scripts/run_results.py all` reproduces every number in this report. Each result also has its own command in the README, and `pytest` runs 50 tests, all passing.

- **Package** `src/keeling/`: one fit/predict class per rung, from `LinearModel` to `ExponentialSeasonalModel`. Parameters are saved as JSON in `results/params/`, and `decisions.md` logs 14 decisions.
- **Tests** include model selection on synthetic data with a known degree and number of harmonics.
- **Tool and demo:** `python scripts/run_tool.py` gives the trend/season split, the growth rate now and the 2035 forecast. The explorer shows the held-out months, then 2035.

10. Contribution statement

Both of us worked on every rung, switched who typed at least every hour, and can explain every line of the code.

Part	Lead	Also worked on it
Data and rungs 1–2		
Rungs 3–4		
Rung 5 and forecast		
Software and tests		
Demo, presentation and report		

Use of AI assistants: we used an AI coding assistant to help write code, the front end and a first draft of this report. We checked every number in it against the output of our own scripts. *(Adjust this to the instructor's rule.)*

Azwad Islam Sachcha _____ Date _____

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